

tenance. The solution can be used collaboratively by customers and partners.

Ideal conditions for quick adoption

It is clear that the industrial metaverse can give enterprises a tangible advantage. We asked our experts what kind of infrastructural developments need to happen in the near future to make conditions ripe for mass adoption of the industrial metaverse.

Garg says, “In the next ten years, metaverse would redefine the internet, much like how mobile technology has redefined communication in the past twenty years. Industrial metaverse would help jump-start many industrial companies from the classical e-commerce model to immersive commerce (i-commerce). For this pace of adoption, progress would have to be made on five dimensions. First, the industrial metaverse experience would need to become device independent so that users can connect to the metaverse with the device of their choice—mobile, laptop, headsets like Oculus, and so forth. Second, users must feel safe in the metaverse and hence identity management and robust transaction methodology using blockchain would need to evolve and become standardised. Third, the creation of a community of experts would need to be fast-tracked and proactively built. Fourth, common standards must evolve to allow seamless navigation between different metaverses. Finally, the wide-scale roll-out of 5G network that promises high bandwidth and low latency would enable a high-quality user experience and act as a force multiplier to the metaverse.”

Kingsley adds, “From a mass

adoption standpoint, the cost of advanced human-machine interfaces and the compute and energy constraints on devices to derive optimum functionality are some of the significant constraints. However, I envision this improving with better abilities, designs, battery life, and long-term usability in the next three years.”

Nashawaty feels that, “A platform needs to be established for early adoption. Education and buy-in would need to be demonstrated to explain why using the industrial metaverse would be an advantage and also an accelerator to future analysis and product/platform enhancements.”

“While the digital twin is at the heart of the industrial metaverse, it alone won’t suffice to fill the metaverse with life. We not only will need a lot more computing power, faster networks, but also will need more powerful artificial intelligence, better chips, and more advanced virtual reality technologies,” remarks Tutt.

Experts



Dale Tutt, Vice President of Industry Strategy, Siemens Digital Industries Software



Hendrik Witt, Chief Product Officer, TeamViewer



Immanuel J. Kingsley, Metaverse Technology Officer and Head of Innovation Lab at Hexaware Technologies



Paul Nashawaty, Senior Lead Analyst, Enterprise Strategy Group - a division of TechTarget



Sadaf Siddiqui, General Manager - Industry Marketing, Keysight Technologies



Umang Garg, MD, Global Practice Leader, Nagarro

The great Indian opportunity

“We see a lot of startups with capable software engineers launching unique solutions across the board. Apart from this, even the front-end companies offering hardware solutions like sensors or connectivity are expanding their presence in India,” says Siddiqui.

He adds that the official launch of 5G services and massive digital enablement projects will definitely push this space in the country from an adoption perspective.

In his interview with YourStory, Jose said, “For industrial applications of metaverse, it needs a confluence of skills—what Flutura

calls the graceful collision of ‘experienced grey-haired folks’ with the ‘young pony tails!’ The experienced grey-haired folks provide domain context to the problem in refining, green energy, process chemicals, and the like. The young pony tails are hungry hackers with limitless energy to find uses for the metaverse and AI to solve real world problems. India is a melting pot of grey-haired folks and pony-tailed folks.” An interesting perspective, indeed!

He explains that India is at the cusp of a renaissance where the country can be No. 1 in industrial AI and industrial metaverse applications, citing examples of startups ranging from Tonbo Imaging and ExactSpace to Flutura and Detect Technologies. Corporations like Tech Mahindra, Wipro, Hexaware, and TCS are also into digital twins and metaverse technologies. As Jose says, “A graceful mix of risk capital, innovative applications, and real-world experience can make the vision of becoming No. 1 a reality through flawless execution.” **EFY**